

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	19 July 2026
Team ID	
Project Name	Nightjar & Co. – MERN BookStore
Maximum Marks	4 Marks

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration using name, email, and password
		Input validation during registration
		Password encryption using bcrypt
FR-2	User Authentication	User login using email and password
		JWT-based authentication
		Protected routes for authenticated users
FR-3	Book Management	Display all books
		View book details
		Search books by title
		Filter books by category
FR-4	Shopping Cart	Add books to cart
		Remove books from cart
		View cart items
FR-5	Admin Book Management	Add new books
		Update existing books
		Delete books
FR-6	User Interface	Responsive design
		Navigation using React Router
		User-friendly interface

Non-functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application provides a simple, intuitive, and responsive interface that enables users to browse, search, and purchase books with minimal effort.
NFR-2	Security	User passwords are encrypted using bcrypt, and authenticated sessions are protected using JWT-based authentication.
NFR-3	Reliability	The system consistently performs user authentication, book retrieval, and cart operations while maintaining data integrity.
NFR-4	Performance	REST APIs are optimized to provide quick responses for browsing books, searching, and user authentication.
NFR-5	Availability	The web application is designed to be accessible through standard web browsers whenever the server is running.
NFR-6	Scalability	The modular MERN architecture allows future enhancements such as payment integration, recommendations, order tracking, and cloud deployment without major redesign.



Figure 1 : Functional and Non-Functional Requirements